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Assignment no. 2

Topic name: Spinal cord and Peripheral Nervous System

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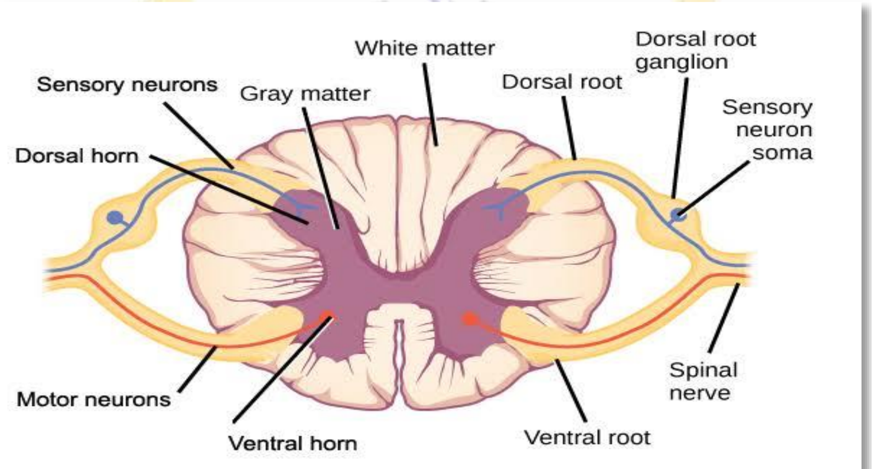
Spinal Cord:

The spinal cord is a part of the central nervous system. It forms a vital link between the brain and the body. It is a long pipe-like structure arising from the medulla oblongata, part of the brain consisting of a collection of nerve fibers, running through the vertebral column of the backbone.

In adults, the spinal cord is usually 40cm long and 2cm wide. Covering the spinal cord and its protective layers is the vertebral column, or the spinal bones. These bones start at the base of the skull and extend down to the sacrum, a bone that fits into the pelvis. The cervical, thoracic, and lumbar regions have different numbers of bones. Most people have seven spinal bones in the cervical column, 12 in the thoracic column, and five in the lumbar column.

Structure of spinal cord:

Cross-section of spinal cord displays:



Cross Section of Spinal Cord

Gray matter:

The gray matter is the dark, butterfly shaped region of the spinal cord made up of nerve cell bodies.

White matter:

The white matter surrounds the gray matter in the spinal cord and contains cells coated in myelin, which makes nerve transmission occur more quickly. Nerve cells in the gray matter are not as heavily coated with myelin.

Posterior root:

The posterior/dorsal root contains afferent nerve fibers, which return sensory information from the trunk and limbs to the CNS.

Anterior root:

The anterior root is the part of the nerve that branches off the front of the spinal column. Anterior roots transmit motor information.

Spinal ganglion:

The spinal ganglion is a cluster of nerve bodies that contain sensory neurons.

Cerebrospinal fluid (CSF):

Surrounds the spinal cord, providing cushioning and protection.

Meninges:

Meanings are three protective layers that surrounds the spinal cord, dura matter, arachnoid matter

- **Dura mater**: This is the outermost layer of the spinal cord's meninges. It is a tough, protective coating.
- **Epidural space**: Between the Dura and Arachnoid space is the epidural space
- **Arachnoid mater**: The arachnoid mater is the middle layer of spinal cord covering.
- **Subarachnoid space**: This is located between the arachnoid mater and pia mater. Cerebrospinal fluid (CSF) is located in this space.
- **Pia mater**: The pia mater is the layer that directly covers the spinal cord.

Spinal nerve:

The posterior and anterior roots come together to create a spinal nerve. There are 31 pairs of spinal nerves. These control sensation in the body, as well as movement. These include:

- Eight **cervical nerve** pairs (nerves starting in your neck and running mostly to your face and head).
- Twelve **thoracic nerve** pairs (nerves in your upper body that extend to your chest, upper back and abdomen).
- Five **lumbar nerve** pairs (nerves in the low back that run to your legs and feet).
- Five **sacral nerve** pairs (nerves in the low back extending into the pelvis).

Function of spinal cord:

The spinal cord plays a vital role in various aspects of the body's functioning. Examples of these key functions include:

- **Carrying signals from the brain**:

The spinal cord receives signals from the brain that control movement and autonomic functions.

- **Carrying information to the brain**:

The spinal cord nerves also transmit messages to the brain from the body, such as sensations of touch, pressure, and pain.

- **Reflex responses**:

 The spinal cord may also act independently of the brain in conducting motor reflexes. One example is the patellar reflex, which causes a person's knee to involuntarily jerk when tapped in a certain spot.

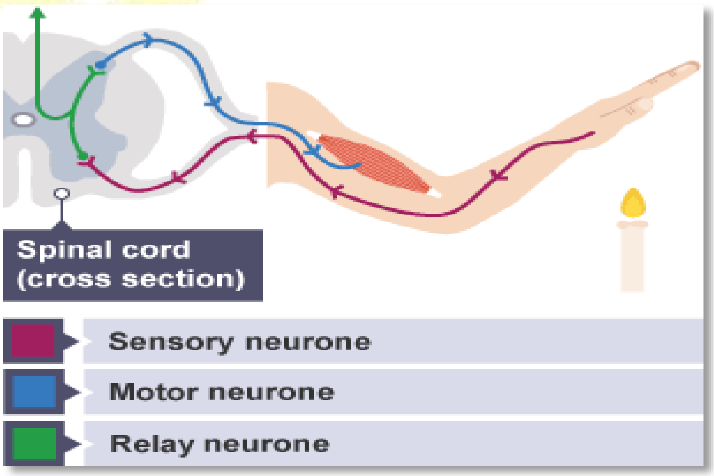
Reflex action or arc:

Reflex is an involuntary and sudden response to stimuli. It happens to be an integral component of the famed survival instinct. In a reflex action, the signals do not route to the brain – instead, it is directed into the synapse in the spinal cord, hence the reaction is almost instantaneous. Events involved in reflex action:

- The first event begins with the receptor detecting a stimulus from a sensory organ. The stimulus could be in the form of pressure, temperature or chemicals.
- This is followed by the sensory neuron sending a signal to the relay neuron.
- The relay neuron then sends the signal to the motor neuron.
- The motor neuron sends a signal to the organ or a cell that acts to the stimuli called the effector.
- Finally, the effector produces an instantaneous response, such as a knee-jerk reaction.

Example:

A simple reflex arc happens if we accidentally touch something hot. Receptor in the skin detects a stimulus (the change in temperature). Sensory neuron sends electrical impulses to a relay neuron, which is located in the spinal cord of the CNS. Relay neurons connect sensory neurons to motor neurons. Motor neuron sends electrical impulses to an effector. Effector produces a response (muscle contracts to move hand away).



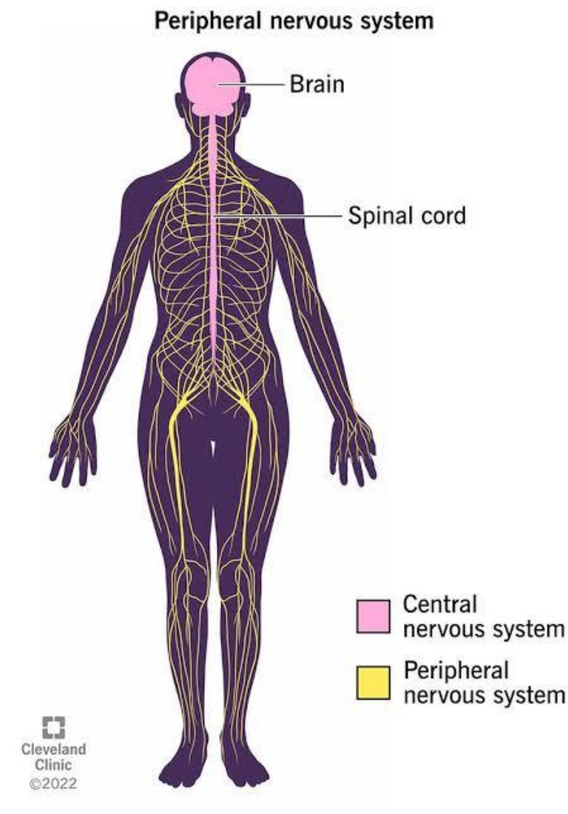
Injuries or damage to the spinal cord can result in various conditions, such as paralysis, spasticity, or sensory loss, depending on the severity and location of the injury.

Peripheral Nervous System

The nervous system is the controlling system of the body and is composed of nerve cells and organs. It is further classified into the central nervous system and the peripheral nervous system.

The central nervous system comprises the brain and the spinal cord. The peripheral nervous system comprises the network of nerves connected to the brain and the spinal cord.

The peripheral nervous system refers to parts of the nervous system outside the brain and spinal cord. It includes the cranial nerves, spinal nerves and their roots and branches, peripheral nerves, and neuromuscular junctions.



It consists of:

1:Cranial Nerves

2:Spinal Nerves

Cranial Nerves:

Cranial nerves are 12 pairs and they emerge from the brain. Some of the examples of cranial nerves are optic, olfactory, etc.

Spinal Nerves:

Spinal nerves have their point of emergence as the spinal cord. There are 31 pairs of spinal nerves. They emerge from the spinal cords into dorsal and ventral roots. At the junction of these two roots, the sensory fibers continue into the dorsal root and the motor fibers into the ventral root.

Peripheral Nervous System Functions:

Following are the important functions of the peripheral nervous system:

- The peripheral nervous system connects the brain and the spinal cord to the rest of the body and the external environment.
- It regulates internal homeostasis.

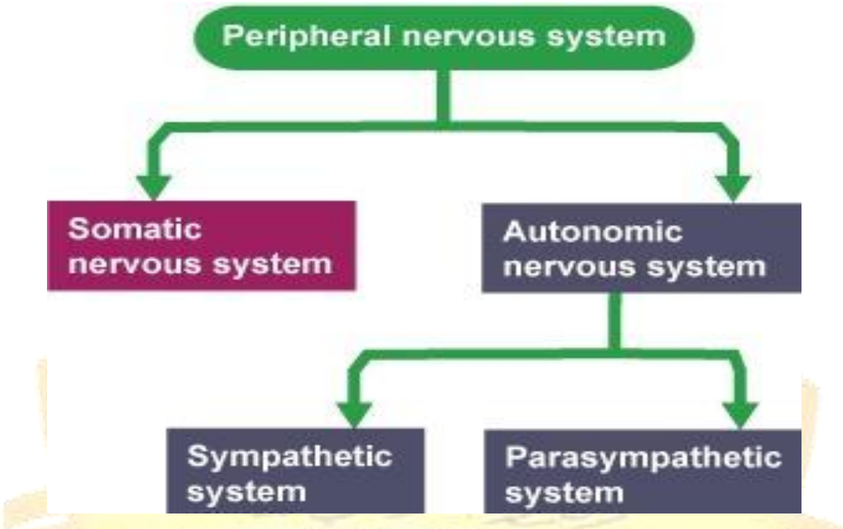
- It can regulate the strength of muscle contractility.
- It controls the release of secretions from most exocrine glands

Division of Peripheral Nervous System

The peripheral nervous system has two divisions:

1. Somatic Nervous System

2. Autonomic Nervous System



Somatic Nervous System

The somatic nervous system connects the central nervous system with the body's muscles and skin. Its primary function is to control voluntary movements and reflex arcs, while also helping us process the senses of touch, sound, taste, and smell.

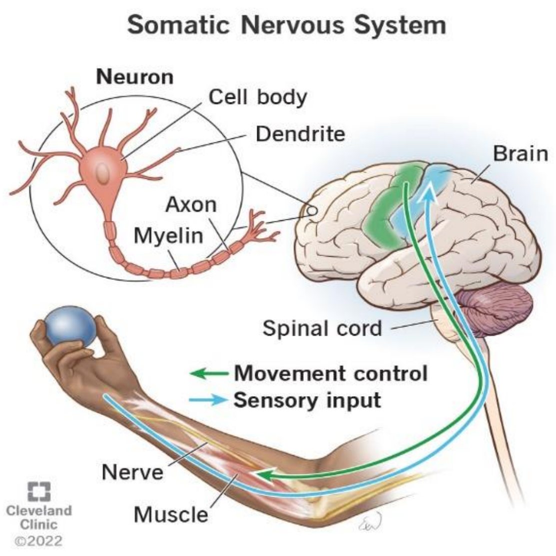
Example:

An example of a somatic system function is if you are out for a jog in the park one brisk winter morning and as you run, you step on a patch of slick ice. Once your foot starts to slip, your somatic nervous system carries a message to the muscles in your legs, enabling you to catch yourself and avoid a fall.

Somatic nervous system has two main jobs:

• **Sensory input:.** All but one of the senses travel through somatic nervous system to reach brain (sight is the exception because retina and optic nerve connect directly to brain). The other senses on head — sound, smell, taste and touch — all use somatic nervous system to reach brain. Sense of touch below your neck uses your somatic nervous system to reach your spinal cord, which then relays signals to your brain.

• **Movement control:** Body's muscles rely on signals that give them instructions to help you move around. The signals from brain must pass through somatic nervous system to reach those muscles and make them move.



Autonomic Nervous System:

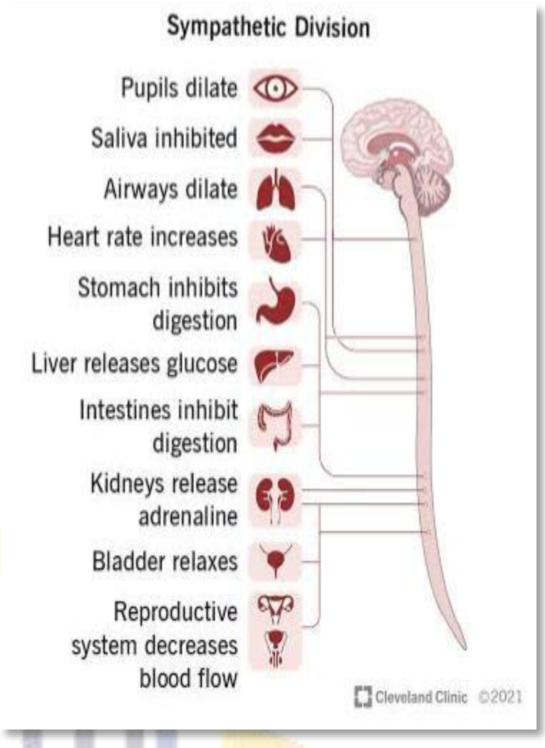
The autonomic nervous system relays impulses from the central nervous system to the involuntary organs and smooth muscles of the body. The autonomic nervous system is a network of nerves that regulates unconscious body processes. The autonomic system is the part of the peripheral nervous system responsible for regulating involuntary body functions, such as heartbeat, blood flow, breathing, and digestion.

It is divided into two parts :

1. Sympathetic Nervous System
2. Parasympathetic Nervous System

Sympathetic Nerves System:

The sympathetic nervous system consists of nerves arising from the spinal cord between the neck and waist region. The sympathetic nervous system's primary process is to stimulate the body's **fight or flight response.** It is, however, constantly active at a basic level to maintain homeostasis. It prepares the body for violent actions against abnormal conditions and is generally stimulated by adrenaline.



The SNS has a major role in various physiological processes such as blood glucose levels, body temperature, cardiac output, and immune system function.

The SNS is responsible for:

1. **Stress response:** Activates the body's response to stress, including increasing heart rate and blood pressure.

2. **Emergency response:** Enables the body to respond quickly to emergency situations.

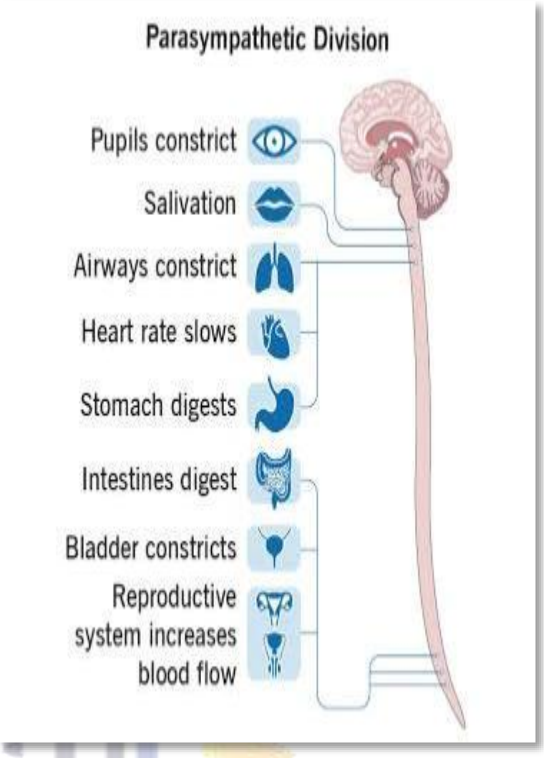
3. **Exercise and physical activity:** Prepares the body for physical activity by increasing heart rate and blood pressure.

4. **Emotional responses**: Involved in emotional responses such as fear, excitement, and anxiety

<u>Parasympathetic Nerves System:</u>

The parasympathetic nervous system is located anterior in the head and neck and posterior in the sacral region. It is mainly involved in the re-establishment of normal conditions when violent action is over. It primarily stimulates the body's "rest and digest" and "feed and breed" responses. It primarily regulates visceral organs. While providing control to many tissues, the parasympathetic system never tries to take control of the maintenance of life.

Parasympathetic Nervous System (PNS) responsible for:



1. **Rest and digest response**: PNS promotes relaxation and digestion.

2. **Regulates sleep**: PNS helps regulate sleep and circadian rhythms.

3. **Lowers heart rate and blood pressure**: PNS slows heart rate and lowers blood pressure.

4. **Increases digestion and nutrient absorption:** PNS promotes digestive processes.

5. **Promotes relaxation and reduces stress:** PNS helps reduce stress and promotes relaxation.

6. **Maintains overall homeostasis:** PNS helps regulate various bodily functions, maintaining overall health and balance.

Sympathetic and parasympathetic nervous system are antagonistic to one another. The sympathetic system activates the "fight or flight" response, while the parasympathetic system activates the "rest and digest response.

Damage to the PNS can result in various conditions, such as neuropathy, numbness, weakness, or paralysis, depending on the location and severity of the injury.

Summary:

The spinal cord and peripheral nervous system (PNS) are essential parts of the nervous system, responsible for:

1. Sensation: Perception of sensory information from the environment.

2. Movement: Voluntary and involuntary movements, including reflexes.

3. Autonomic functions: Regulation of bodily functions, such as heart rate, blood pressure, and digestion.

4. Reflexes: Rapid responses to stimuli, without conscious thought.

5. Communication: Transmission of signals between the central nervous system (CNS) and the rest of the body.